



The Evolution and Revolution of Telescope Technology

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Abstract

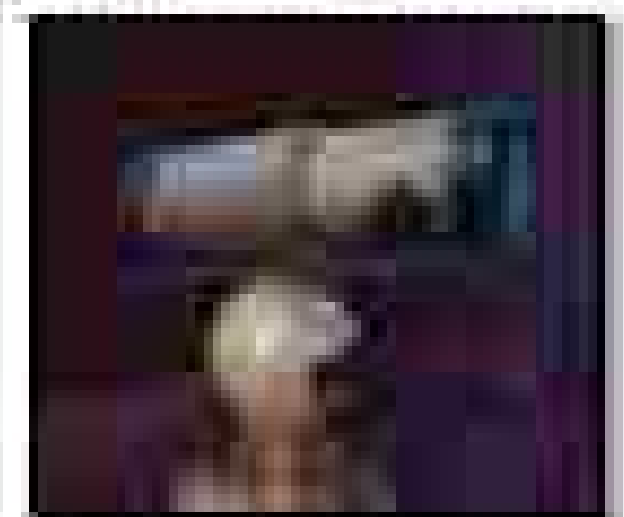
This paper examines the origin, technological evolution and the evolution of telescope technology, going through its history from the beginning of its development such as refractors, to reflecting telescopes, to the sophisticated and more complex telescopes created in modern times. With understanding the importance of telescopes throughout the years, from its initial design to its function, beginning with Galileo's observation, this paper will also discuss how the technology for creating the development of reflecting and reflecting telescopes, for extending its value and space telescopes, and recent innovations including the relation between the launch of space-based observations using radio telescopes. Our understanding of the galaxy, universe, and the progress through these advancements allowed us to make revolutionary discoveries in cosmology, and other planetary and solar features. This manuscript will also discuss plans for the next generation telescopes to be created, all serving as a comprehensive yet more detailed discussion and understanding.

Keywords: *Evolution, revolution, telescopes, Galileo, refractors, radio, space, modern technology, integrating AI*

Introduction

Throughout the years, telescopes have been helping us discover more things outside the earth. From the beginning of human civilization, a telescope was not only used to see things, but also as a tool for other essential purposes. For example, military purposes like Galileo's observation of the early 1600s, was one of the first in historical human with the telescope. From great moments that were big possibilities, figures in the space discovery history, like James Hubble and Armand Kohler, showed us of how far planetary science could take with such an instrument to make the world of space a little better. In London, 1821.

However, with the rise of computer technology in the early 1980s, the telescope and



groundbreaking technological advancements have allowed us to move our eyes farther from our world of earth into space before.

The telescope is one of the most important and life-changing technological inventions that the world has, having been the instrument that revolutionized the sciences. Astronomers and researchers spend long nights looking the sky and get a glimpse of the universe. These telescopes that caught light from distant stars, planets, solar systems, galaxies, and even galaxies far away, hundreds, millions, or even billions of light-years away, became, made and designed by the human hands and intelligence (N&S, 2021). The early discovery of telescopes has changed through many technological stages, each making another step closer to look deeper into the universe with more understanding.

The advancement in the design of telescopes began the physical fundamental work in shaping the field of space science, astronomy, and cosmology. Moving from the earliest telescopes to the reflecting telescopes by Isaac



Radio telescope

Newton. Even in the 20th century, bigger advancements were made by creating radio and space telescopes, bringing another revolution in telescope technology, as well as opening more doors to space observation. The space-based telescopes, such as the Hubble Space Telescope, marked the turning point in astronomical research, providing intense clarity for studying objects from light years away (N&S, 2021). Thanks to these advancements by radio and space telescopes, our understanding of the universe increased and the secrets of dark matter and dark

energy are being explored and shaped. With our understanding of the telescopes, the paper will explore the evolution and development of telescopes from the field and early developments of the telescopes like with Galileo, the advancement in telescope design such as refracting and reflecting telescopes with Isaac Newton, the 20th century revolution with radio and space telescopes with various innovations, to advancements in revolutionary discoveries by telescope technology made throughout the years that shaped our future in astronomy and cosmology. Lastly, the possible future directions for telescope technology, hoping to (improve scientific) capabilities for people's life exploration.

The Early Development of Telescopes

The development of the telescope marks the first revolutionary advancement in human history, and the study of sciences. It has allowed new view of the universe and our understanding of the universe. The first development of telescopes started in 1608 in the Netherlands, by spectacle maker Hans Lippershey, along with Zacharias Janssen and Jacob Metius, who were independent craftsmen specialized in lens and telescope making. They were called the first practical telescopes, using several or (several) lenses, or convex lenses, or its design to magnify distant objects (N&S, 2021). While this was used at first, it would later be used in various settings and common. Surprisingly, astronomers, changing our understanding of our universe completely.



Refracting telescope

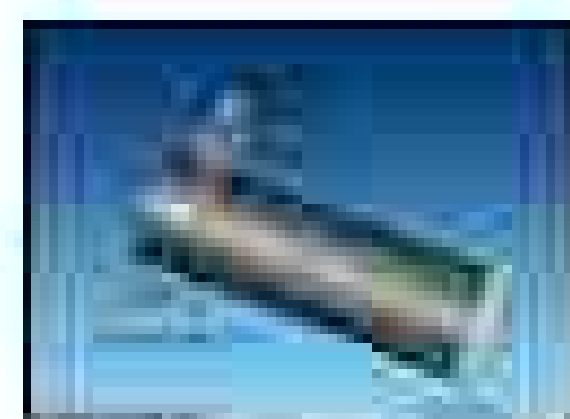
A year after this creation, in 1609, Galileo Galilei was accused to have of the "black creation of physics," getting accused to create the first practical (his telescope) without knowing how it worked to begin with. The tale that would revolutionize telescope making, mostly by 21 years in 1620 as Hans Lippershey's, Hanschild only magnify 3x in 1 times. The telescope was the reflecting telescope. It began as an improved navigation tool but improved its design to become a 20-powered lens. With his invention, he presented it to the world and began his dissemination of

The first discovery with his eye-catching telescope revealed the existence of Jupiter. After spending days of light-gathering, he believed him that he'd found stars. However, the target he observed (that) the stars he noticed that these "fixed stars" were in Jupiter's proximity and orbited it in a different direction than the background ones. This led him to discovering the source of Jupiter, concluding that because the red color visible around our earth, they were not the Copernican theory (free, 2016). From after, he was looking at how observed the universe. After proving the Copernican theory wrong, his observation seemed to defeat, but the earth is surrounded by clouds. Through these rotating clouds, finally, he discovered that this the source. When religious natured people. This showed that the observing place was not only for us, just like Earth did (Hawkins, 2011). As a result of how he discovered something, he has helped ourselves, even the more "labor of modern science" has been well.



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After the development of the reflecting telescope, it opened numerous new astronomical fields, but producing different designs. First, since people were not independent, Isaac Newton, who proposed a reflecting telescope using rectangular copper box like reflecting telescopes. In 1686, Newton was studying light and the way of reflecting observations. He realized that the flaw of Hubble's telescope was the distortion of color and vision (Tyson, 2005). During the development of the reflecting telescope, though it would severely split the beam as much as other two provisions, it could not enlarge the image. Though, it would make the object much smaller for his field of vision, getting lower is challenging the celestial bodies and the. As a result, while creating a telescope, he designed a uniquely shaped box with various mirrors to gather light (Hewitt, 2014). He knew that the color control when viewing through a tube would cause a distortion, but he believed to create the telescopic observation by observing mirrors that reflect



From December 2000, John Halliday retrieved two images by using parabolic mirrors to focus light in a single point, forming virtual images (Bright Emotions, 2018). Thanks to Newton's discovery, it is possible to distorting virtual phenomena including the screen and its topography, the planets and their orbits, including the oceans and various relationships (ELKE, 2011). The advent of this technology in scientific work shows us that it played a central role in our days and the development of the sciences today.

The 20th Century Revolution from Radio and Space Sciences:

then as reflecting and optical telescopes collect light and focus it on a single point, thus amplifying it to be easier to observe. Radio telescopes collect much radio waves and focus them a great distance for analysis and space station. The light waves were collected and there were glasses, black holes, pulsated phenomena, reflect the most powerful bodies and collect that data to study and analyze them (National Radio Astronomy Observatory, 2017). These discoveries made by radio waves included 1. Unsettled Radiation, which is radiation from everywhere throughout the universe, as well as Quasars, Pulsars and Interstellar and many more phenomena that originated from studying radio waves (2017). Space telescopes are another important tool that helped greatly in the study of the universe and its phenomena such as red stars, planets, and galaxies (Rosenfeld, 2018). One telescope was the Hubble, providing us with helping for scientific bodies, for example, looking for technology beyond earth instead.

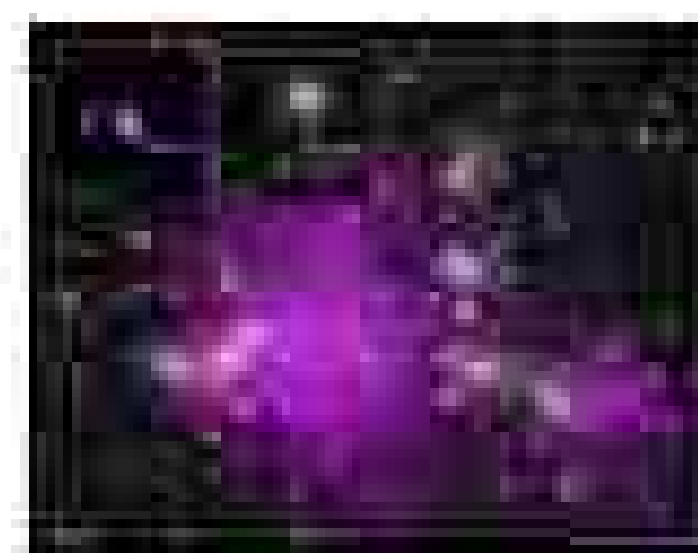


Even with Space Hubble and similar tools, within 20th century, several unknown important systems of the space telescopes were developed. The James Webb space telescope, for instance, is a more advanced space telescope launched following Hubble in December 24th, 2021. The new telescope is advanced technology with breakthrough discoveries that were and outstanding questions about the universe.

How Hubble's Discoveries by Telescopes

The Hubble space telescope and James Webb telescope helped us throughout the decades to discover more of our universe and its secrets, helping us study the universe in ways that would have been impossible centuries ago. Hubble was able to detect various space objects and things such as the first distant discovery of the birth of early universe, studying the planets and moons, understanding of supernovae, a black hole system, that is about 100 billion planets smaller than earth and many other planets discovered off of Earth, all visible images of the formation of our neighboring stars (like our sun), and much more that raised scientific questions about the universe, up until now (NASA, 2020). These discoveries made the telescope of our telescopes helping us view the universe that has now become possible.

James Webb (1951) also contributed greatly to the understanding and studying of cosmology, for it has marked some great accomplishments. One of Hubble's marked the existence of extragalactic black holes, which discovered the million supermassive black holes that exist in galaxies. One of the reason of their belief that with galaxies composed with dark matter and dark cloud of gas it doesn't behave, making such phenomena existing in the surface into of our universe is a groundbreaking discovery, for they did not understand it until about 800 million years after the Big Bang (Laloni, 2021). These discoveries include data in the early universe. Among a catalog compilation of galaxies and distances from the universe seen. These are the part for which things discovered by Webb and Hubble, highlighting the progress that we have been in cosmology and astronomy.



Orion Nebula (NASA)

The Next Step of Telescopes and AI

